Week 5 Make a rule work · paper route answers

Teacher only. Keep this sheet away from pupil copies.

TEACHER ANSWERS - WEEK 5 PAPER ROUTE. Every artefact, with the model answer and with what counts as correct when a pupil points, says or draws instead of writing.

WHY THE PAPER COUNTER IS EXACT, NOT AN APPROXIMATION. The real Player sprite is 16 stage units across. The paper counter is one square, 10 units across. This was checked, not assumed: at every one of the 1,813 positions on the 10-unit grid, the 16-unit sprite and the 10-unit counter give the SAME touching answer against this maze. This holds because every wall edge in the maze is a multiple of 10, so both sizes reduce to the same rule - touching if the centre is on, or between, the two edge numbers of a grey band. So a pupil tracing on paper and a pupil running on screen get identical results. If you ever change the maze so a wall edge is not a multiple of 10, this stops being true and must be re-checked.

P1 GRID 1 - THE WALL RULE (the new Week 5 script). Model answer, in order under the hat: row 1: forever / row 2: if then, with touching [Maze]? in the question space / row 3: go to x [start x] y [start y], written inside if then / rows 4 and 5 left empty. The two spare rows staying empty is correct and is worth saying so out loud - not every space has to be filled. Correct nesting looks like: forever holds the if then, and the if then holds the go to. Accept indents, drawn boxes, arrows, or the pupil putting a finger on each space and telling you which block belongs there. WRONG ANSWERS TO WATCH FOR: go to written after the if then instead of inside it (this is the misconception in the pack - Player would return even in clear space); touching [Finish]? instead of touching [Maze]?; a change x or change y block put in this script (those belong in the four key scripts, and the word bank includes them on purpose so the choice is real).

P1 GRIDS 2 to 5 - THE FOUR KEY SCRIPTS. right arrow: change x by 10. left arrow: change x by -10. up arrow: change y by 10. down arrow: change y by -10. One block each, two spare rows empty. The minus signs matter. If a pupil writes change x by 10 under the left arrow, do not correct it yet - send them to the grid with the counter and let the walk show them.

P1 GRID 6 - THE START SCRIPT. go to x -180 y -120, or the pupil's own chosen clear start written as go to x [their x] y [their y]. Two spare rows empty. This is a SECOND green flag hat. If a pupil joins it onto the wall rule to make one script, that is a real and common error - ask them which block runs first and whether the go to should happen once or every time round the loop.

P1 GRID 7 - THE BROKEN EXAMPLE. What the pupil writes: row 1: if then, with touching [Maze]? in the question / row 2: go to x -180 y -120, inside the if then / rows 3 and 4 empty. THEN the pupil rings or points to where forever should be -

directly under the hat, above the if then. Correct by pointing: the pupil puts a finger in the space between the hat block and the if then and says forever, or draws an arrow into that space. Do not accept 'it is wrong' with no location. The whole point is knowing WHERE the missing block goes.

P2 - THE MAZE GRID. The start x -180, y -120 is clear: nearest grey is the left wall, whose inner edge is at x -200, two squares away. The touching rule the pupil says aloud - 'any part of my counter on grey is touching' - is exactly equivalent to the Scratch touching block on this maze. Tie-breaker if a pupil is unsure at an edge: the counter's centre dot sitting ON a grey band's edge line counts as touching. Both rules give the same answer everywhere here.

P3 TABLE 1 - PREDICT. x 0, y 0: NO, not touching, nothing runs, the counter stays. x -60, y 0: YES, touching, go to runs, the counter returns to the start. x 60, y 0: YES, touching, go to runs, the counter returns to the start. x -180, y -120: NO, not touching, nothing runs. These are the same three places as the screen lesson's Test clear space, Test wall 1 and Test wall 2 buttons, and they give the same answers. A wrong prediction that the pupil then corrects after looking is a BETTER outcome than a right prediction - the prediction box is not marked, the change of mind is the evidence. Pointing counts: a pupil who points to a Yes card and a No card is answering.

P3 TABLE 2 - TRACE 1, THE CLEAR WALK. Step 2: x -180, y -120, No, counter stays. Step 3: x -180, y -110. Step 4: x -180, y -110, No, counter stays. Step 5: x -180, y -100. Step 6: x -180, y -100, No, counter stays. Step 7: x -180, y -90. Step 8: x -180, y -90, No, counter stays. Note x never changes - only y. A pupil changing x here has used the wrong key script. In the 'where the counter goes' boxes accept 'stays', 'nothing', a dash, or the pupil leaving the counter still and saying nothing happens.

P3 TABLE 3 - TRACE 2, WALL ONE. Step 2: x -180, y -120, No, counter stays. Step 3: x -190, y -120. Step 4: x -190, y -120, NO - this is the one that catches people out, x -190 is still clear, the counter's edge stops just short of the wall. Step 5: x -200, y -120. Step 6: x -200, y -120, YES. Step 7: x -180, y -120 - the go to block runs and the counter slides back to the start. Step 8: x -180, y -120, No - the loop checks again and now the answer is No, so nothing runs and the counter rests. Step 8 is the step that shows the loop did not stop. Do not skip it.

P3 TABLE 4 - TRACE 3, WALL TWO. Step 2: x 40, y 0, No, counter stays. Step 3: x 50, y 0. Step 4: x 50, y 0, YES. Step 5: x -180, y -120 - the counter returns to the start. Step 6: x -180, y -120, No, nothing runs. This is a DIFFERENT wall from Trace 2, which is what outcome 3 asks for. If a pupil only ever tests the left wall, outcome 3 is not evidenced - send them to this one.

P3 TABLE 5 - COMPARE. Row 1: broken - No. Mine - No. Row 2: broken - NO, it does not check here, the check already happened at the green flag and will not happen again. Mine - YES, forever checks here. Row 3: broken - No, still no check. Mine - Yes, and this one is touching. Row 4: broken - the counter finishes sitting at x -200, y -120, inside the wall, stuck. Mine - the counter finishes at x -180, y -120, back at the start. Row 5: mine, the one with forever. Accept the pupil placing two counters on the grid, one for each script, and pointing at where each ends up - that is the clearest possible answer and needs no writing at all. THIS TABLE IS THE MAIN EVIDENCE OF UNDERSTANDING REPETITION. Watch the pupil do it; do not just collect it.

P3 TABLE 6 - CHALLENGE. Row 1 prediction: Player will be stuck. Row 2: touching at x 50 y 0, so go to runs, and the counter goes to x -60, y 0. Row 3: x -60, y 0 is INSIDE the left inside wall, so the very next check is Yes again, so go to runs again, and the counter is sent to x -60, y 0 again - and again, and again. The counter never gets free. This is the fault: a return place inside a wall traps Player forever. Row 4: repair by choosing a start with clear squares on every side - x -180, y -120 works, and every go to block on every sheet must be changed to match. Row 5: x -200, y -140 is a corner where the left wall and the bottom wall meet - touching, and the counter returns. Row 6: x 50, y 0 is an ordinary wall - touching, and the counter returns. Both must return to the REPAIRED start. This matches the pack's expected answer 4.

P3 TABLE 7 - MY SAFE START. The model start x -180, y -120: no part of the counter is on grey; there is a clear square on every side (the nearest grey is two squares left, and two squares down). A pupil's own start is correct if it passes both tests - place the counter and look at all four neighbours. Common bad choices: anywhere in the bottom-left corner tighter than x -190 y -130; anywhere between x -70 and x -50; anywhere between x 50 and x 70.

EXPECTED ANSWERS 1 to 5 FROM THE TEACHER GUIDE, ANSWERED ON PAPER. (1) A one-off check runs only when it is triggered. If Player reaches a wall later, the script does not look again. Forever keeps looking during play. The paper proof is Compare row 2. (2) The script belongs on Player, and the sensing block must name Maze. Testing touching Finish would check the wrong object. The paper proof is the sprite heading on grid 1 and the block the pupil wrote in the question space. (3) With no wall contact the condition is false, so the go to does not run. With contact it runs. The paper proof is every No row and every Yes row of Traces 1 to 3. (4) If the reset point is inside a wall, Player keeps returning to the blocked point. Move the reset to a clear space and update EVERY start block to match. The paper proof is Challenge rows 2 and 3. (5) The model start is x -180, y -120. The probe places x -60, y 0 and x 60, y 0 overlap different walls. The clear probe x 0, y 0 has no contact. The paper proof is Predict table row 1 to 3.

IF A PUPIL POINTS, SAYS OR DRAWS RATHER THAN WRITING - WHAT CORRECT LOOKS LIKE. Building the script: the pupil places or points at blocks in the right order and shows the two nesting spaces with a finger. Correct if the order and the two spaces are right. Tracing: the pupil moves the counter and says the coordinates, or reads them off the printed axis labels, and an adult writes them in the boxes. Correct if the pupil chose the move and named the place - an adult may hold the pen, never the decision. Touching: the pupil says Yes or No, or points at a Yes card and a No card, or nods and shakes. Correct if the answer matches the grid. The consequence: the pupil slides the counter back to the start themselves, or tells the adult to slide it. Correct either way - directing an adult is an agreed access method under this pack and must be recorded as the method used, not as a prompt. Drawing: a pupil who draws the counter in two places with an arrow between them has recorded a trace row. Accept it.

WHAT WOULD MAKE THIS NOT EVIDENCE OUTCOME 3, so you can avoid it: giving the pupil a completed grid to copy; tracing FOR the pupil while they watch; testing the same wall twice and calling it two positions; filling in the trace tables from the answers on this page; leaving the prompt boxes blank. Any of those and the observation does not support the Summary sheet.

FINALLY: record the completed outcome on the AQA Summary sheet from your own observation. Do not write on, change or reprint AQA's sheet in any other way. The fact that this pupil took the paper route is recorded in the Teacher Guide only. Outcomes 5 and 6 remain in Weeks 7 and 8 - check every outcome individually.

What this week evidences

AQA 71638 outcome 3: Observe the pupil constructing the collision script, then demonstrate contact at two wall positions. Record the actual consequence and any prompts.

Known limit of the paper route

ONE GAP, NAMED PLAINLY.

On a screen, the forever loop makes the COMPUTER repeat the check by itself, many times a second, with the pupil doing nothing. On paper, the PUPIL repeats the check. So the paper route evidences that the pupil understands repeated checking. It does not show a machine performing that repetition.

Why this matters in practice: a pupil could fill in a trace table correctly and still believe the check happens only once, because on paper they are told when to check.

Two things in this route close that gap, and both are required, not optional.
First, the Compare table. The pupil traces the broken one-check example and their own forever script from the SAME counter walk, and writes the two different endings. A pupil who thinks the check happens once cannot produce two different endings and explain why.
Second, the observation itself. The teacher must watch whether the pupil goes back up to the 'if touching Maze?' row without being told, and must write 'unprompted' or 'prompted' in the prompt box. Write the word. Do not leave it blank. That single word is what separates evidenced understanding from a neatly filled table.

ALSO PLAINLY, so nobody is surprised later: the paper route produces no .sb3 file. The pack check 'I saved and reopened the working rule' is met here by the pupil writing their name and the date on their sheets, filing them, later finding their own sheets without help, and re-tracing one wall test from their own writing. That is a genuine save and reopen of the pupil's own work, and AQA outcome 3 does not require a file - the published evidence requirement is a Summary sheet completed by the teacher from individual observation. But if your centre has separately told a moderator that a digital artefact will exist for every pupil, paper will not produce one. Settle that with the UAS lead BEFORE the pupil starts this route, not after.